Peripherals and connectivity

Q-1 Connect five different types of peripherals (one each from input, output, storage, and communication categories) to your PC and document the connection method (USB, Bluetooth, Ethernet, etc.), including any visible device name in Device Manager or System Preferences.

Hint: Example peripherals: USB mouse, external hard drive, headphones, network adapter, webcam.

ANS: Connecting and Documenting PC Peripherals

Peripheral Details

No.	Peripheral	Category	Connection Method	Example Device Name
1	USB Mouse	Input	USB	HID-compliant mouse
2	Webcam	Input	USB	USB Camera
3	Headphones	Output	Bluetooth	Bluetooth Headphones
4	External HDD	Storage	USB 3.0	USB Mass Storage Device
5	Network Adapter	Communication	Ethernet	Realtek PCIe GbE Family Controller

Steps

1. Connect the **USB mouse** to a USB port and check that the cursor moves.
2. Connect the **webcam** through USB and check it using the Camera application.
3. Turn on the **Bluetooth headphones** and pair them with the PC through **Settings → Bluetooth & devices**. Windows supports Bluetooth peripherals such as headphones, mice, and keyboards.
4. Connect the **external HDD** to a USB port and check whether it appears in File Explorer.
5. Connect the **Ethernet cable** to the network adapter and check the network connection.
6. Open **Device Manager** by searching for "Device Manager" in Windows. Device Manager can show devices by category, such as Network adapters, and provides device status and driver information.
7. Find each connected device and record its visible device name.

Q-2 Install drivers for a USB device manually by downloading the correct driver from the manufacturer's website, and then verify the device is working correctly in Device Manager (Windows) or System Information (Mac).

ANS: Manual USB Device Driver Installation

Device Used

USB Device: USB Webcam
Connection: USB
Operating System: Windows 11

Steps

1. Connect the USB webcam to the computer.
2. Note the **manufacturer and model number** of the webcam.
3. Open the manufacturer's official website.
4. Go to the **Support/Downloads/Drivers** section.
5. Search for the exact webcam model.
6. Download the driver that matches the Windows version and system type.
7. Open the downloaded driver file and follow the installation instructions.
8. Restart the computer if the installer asks for it.
9. Open **Device Manager** by searching for it from the Start menu.
10. Expand **Cameras** or **Imaging devices** and find the webcam.
11. Right-click the webcam and select **Properties**.
12. Open the **Driver** tab and check the driver information.
13. Check the **Device status** under the General tab.

Verification Result

Item	**Result**
Device	USB Webcam
Connection	USB
Device Manager Category	Cameras
Device Name	USB Camera
Driver	Installed
Device Status	This device is working properly
Warning/Error	None

Q-3 Set up a network printer (not USB) using its IP address on your local Wi-Fi or LAN, install the correct printer driver, and print a test page to confirm connectivity.

ANS: Setting Up a Network Printer Using IP Address

Device Used

- **Printer:** HP LaserJet Pro
- **Connection:** Wi-Fi
- **Example IP Address:** 192.168.1.50
- **Computer:** Windows 11

Steps

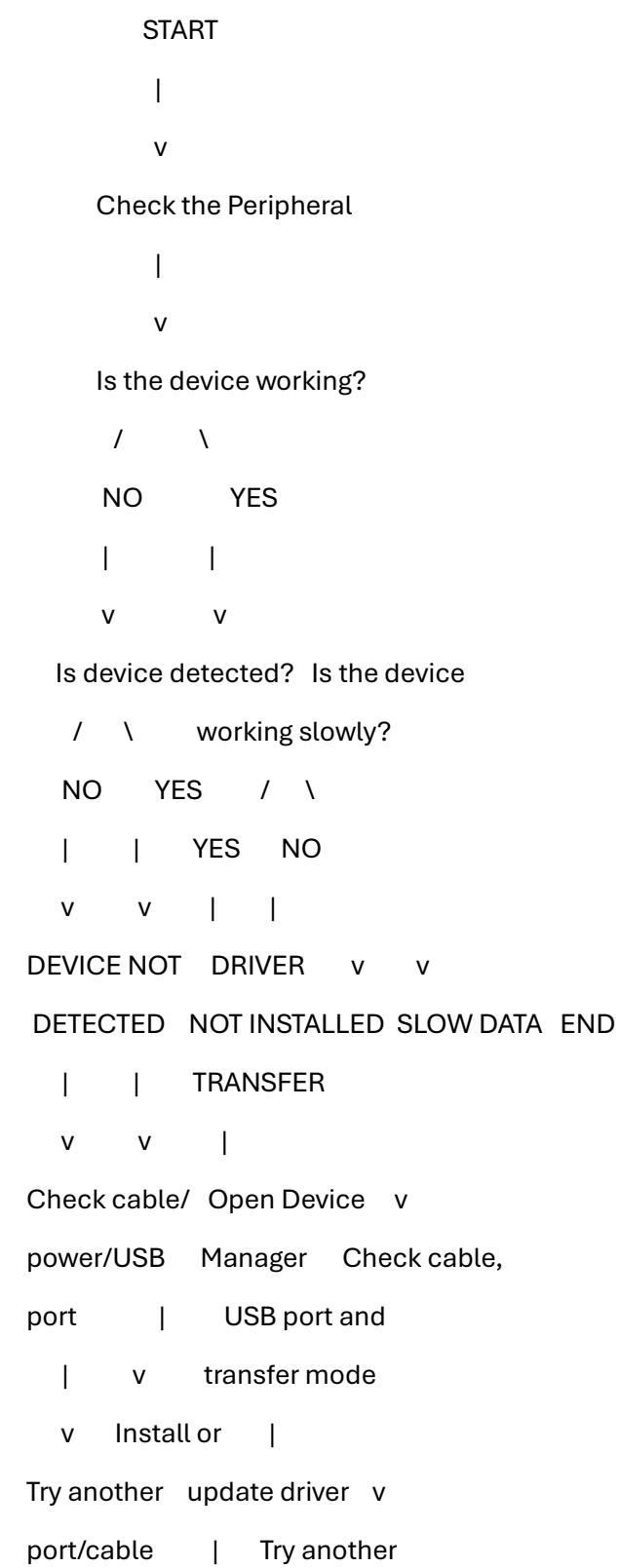
1. Turn on the network printer.
2. Connect the printer and computer to the **same Wi-Fi/LAN network**.
3. Find the printer's IP address from the printer's network settings or network configuration page.
4. On the PC, open **Settings → Bluetooth & devices → Printers & scanners**.
5. Click **Add device**.
6. If the printer is not automatically found, select the option to **add a printer using an IP address or hostname**.
7. Enter the printer's IP address, for example **192.168.1.50**.
8. Download and install the correct printer driver from the manufacturer's official website.
9. Select the correct printer model during installation.
10. Complete the installation and open the printer's **Printer Properties**.
11. Click **Print Test Page**.
12. Check whether the test page prints successfully.

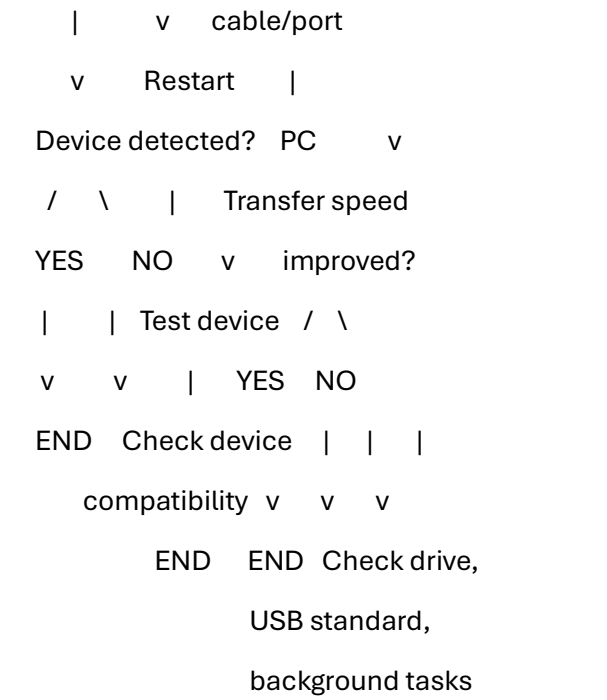
Item	**Result**
Printer	HP LaserJet Pro
Connection	Wi-Fi
IP Address	192.168.1.50 *(example)*
Driver	Correct manufacturer driver installed
Printer Status	Ready/Online
Test Page	Printed successfully
Connectivity	Working

Q-4 Create a troubleshooting flowchart (hand-drawn or digital) that covers common peripheral issues such as 'Device not detected', 'Driver not installed', 'Printer not printing', and 'Slow data transfer', showing the steps to diagnose and solve each problem.

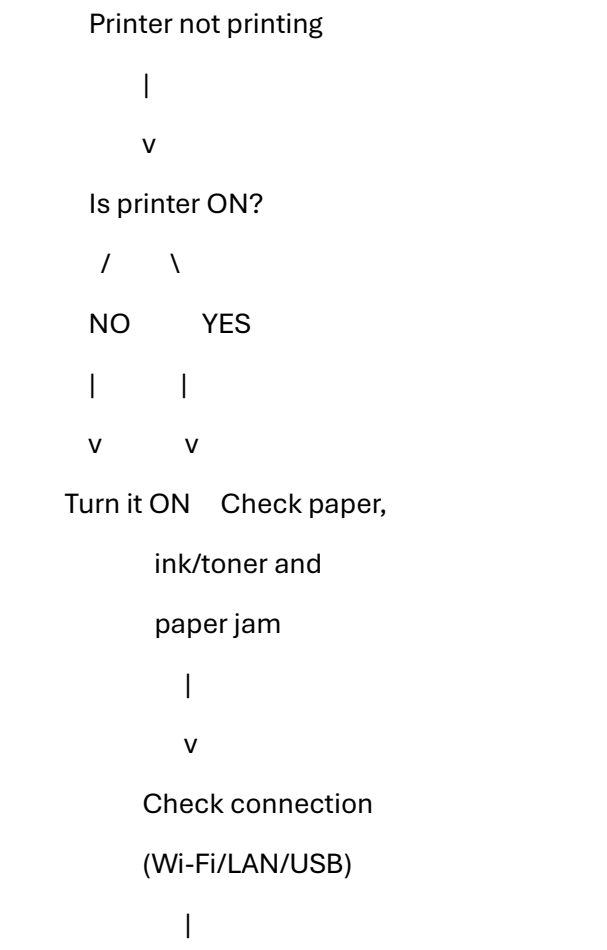
ANS: PERIPHERAL TROUBLESHOOTING FLOWCHART

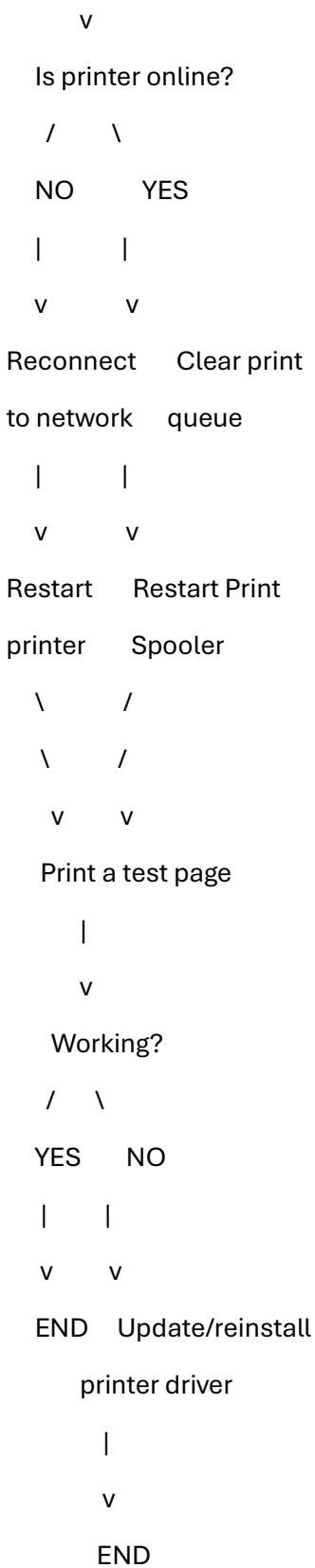
Flowchart:





PRINTER NOT PRINTING





Troubleshooting Steps:

1. Device Not Detected

- Check whether the device is powered on.

- Check the cable or wireless connection.
- Try another USB port or cable.
- Check Device Manager.
- If still not detected, check device compatibility or hardware condition.
- Windows can also allow a device to be added manually when it is not detected automatically.

2. Driver Not Installed

- Open Device Manager.
- Find the device with a warning symbol.
- Right-click and select “Update driver”.
- Install the correct driver from the manufacturer if required.
- Restart the PC.
- Test the device again.

3. Printer Not Printing

- Check printer power, paper, ink/toner, and paper jams.
- Check the Wi-Fi/LAN connection.
- Make sure the printer is online.
- Clear stuck print jobs.
- Restart the Print Spooler service.
- Update or reinstall the printer driver.
- Print a test page to confirm that the printer works. These are also recommended Windows troubleshooting steps.

4. Slow Data Transfer

- Check the USB/data cable.
- Try another USB port.
- Check whether the device supports USB 2.0, USB 3.x, or a faster standard.
- Close applications running heavy background transfers.
- Test with another cable or port.
- If the speed is still low, check the storage device for errors or performance problems.

Q- 5 Design a printer management plan for a startup office of 50 employees: specify how you would organize network printers, manage driver installations, and ensure everyone has access without printer queue conflicts.

Hint: Think about printer sharing, access permissions, and using printer queues or print servers.

ANS: PRINTER MANAGEMENT PLAN FOR A 50-EMPLOYEE STARTUP OFFICE

1. Objective

The main goal is to provide reliable and easy printing for all 50 employees while avoiding printer queue conflicts, incorrect drivers, and unauthorized access.

2. Printer Organization

I would use 3 network printers:

Printer	**Location**	**Main Users**	**Connection**
Printer 1	Administration	HR, Admin, Management	LAN/Wi-Fi
Printer 2	General Office	Most employees	LAN
Printer 3	Accounts/Finance	Accounts and Management	LAN

Each printer would have a fixed IP address and a clear name such as:

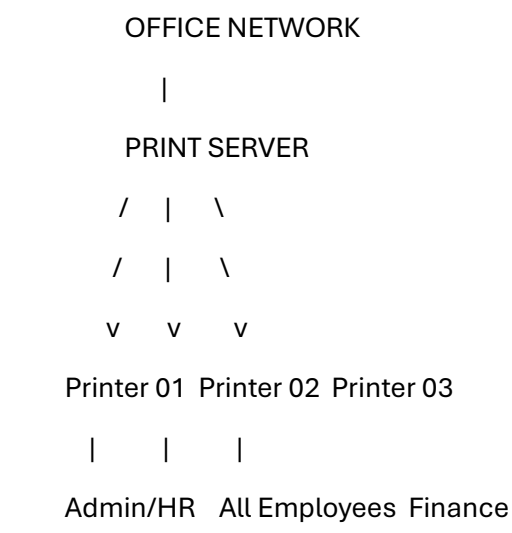
- Office-Printer-01
- Office-Printer-02
- Finance-Printer-01

The printers would be connected to the same office network.

3. Use of a Central Print Server

A dedicated print server would manage all three printers.

Basic setup:



The print server would contain a separate print queue for each printer. Microsoft describes a shared print queue as the central object through which clients send jobs to a printer.

4. Printer Driver Management

- Install the correct manufacturer driver on the print server.

- Use the same approved driver version for each printer model.
- Keep a record of the printer model and driver version.
- Update drivers centrally when a new stable driver is required.
- Employees should not install random printer drivers themselves.
- Where supported, Point and Print can provide the required driver and printer configuration from the print server to the client.

Example:

Administrator

|

v

Install/Update Driver

|

v

Print Server

|

v

Employees connect to

shared printer

5. User Access and Permissions

Employees would be divided into groups.

User Group	**Printer Access**	**Permission**
General Employees	Office-Printer-02	Print
HR/Admin	Printer-01 and Printer-02	Print
Accounts	Printer-03	Print
Management	All required printers	Print
IT Administrator	All printers	Manage Printers/Queues

Normal employees would receive only the Print permission. IT administrators would manage printers, drivers, and print jobs. Windows print permissions can separately control printing, document management, and printer management.

6. Preventing Printer Queue Conflicts

Each physical printer would have its own dedicated queue.

Example:

Office-Printer-01

|

v

Queue 01

|

Employee Jobs

|

v

Printer

Office-Printer-02

|

v

Queue 02

|

Employee Jobs

|

v

Printer

If many employees send documents to the same printer, their jobs will wait in that printer's queue instead of interfering with another printer's queue.

IT staff can monitor the queues and remove stuck or failed jobs when necessary.

7. Printer Naming and Identification

Every printer would have:

- Unique printer name
- Fixed IP address
- Physical location
- Printer model
- Assigned user group

- Driver version

Example:

Printer Name	IP Address	Location
Office-Printer-01	192.168.1.50	Admin
Office-Printer-02	192.168.1.51	General Office
Finance-Printer-01	192.168.1.52	Accounts

These IP addresses are examples and should be replaced with the actual office printer IP addresses.

8. Backup and Maintenance

- Keep one printer available as a backup for important documents.
- Check printer status regularly.
- Keep spare toner/ink and paper.
- Clear stuck print jobs.
- Update drivers when required.
- Keep printer firmware updated according to the manufacturer's recommendations.
- Restrict printer management to IT staff.
- For confidential documents, place the printer in a controlled area and use appropriate access controls. Microsoft also recommends considering physical and data security for printers handling confidential information.

9. Employee Printing Process

Employee

|

v

Select Shared Printer

|

v

Send Print Job

|

v

Print Server

|

v

Correct Printer Queue

|

v

Printer

|

v

Printed Document

Employees only need to select the appropriate shared printer. They do not need to configure the physical printer or install an unrelated driver manually.

10. Result

This printer management plan provides:

- Central printer management
- Controlled driver installation
- Separate queues for each printer
- Easy access for all 50 employees
- Group-based printer permissions
- Reduced queue conflicts
- Easier troubleshooting and maintenance
- Better security for confidential printing